

# The Closed-Loop Gym:

A Brief Technical Note on Why Men Aged 20–27 Strength-Train Through Modernity

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May 2026

**Abstract.** A significant fraction of men aged roughly 20–27 train four to six days per week despite not competing in sport, not performing physical labor, and not, by their own report, being especially unhealthy. This note argues that the gym functions for this cohort less as a fitness facility and more as a *closed-loop control system*: a rare modern environment in which input maps reliably to output, identity is concrete, and progress is publicly legible. We model the gym’s appeal as the product of high effort–outcome correlation  $\rho$  with a society offering vanishing legibility elsewhere—algorithmic dating, AI-disrupted careers, deteriorating young-male social networks. Under that model, the gym is not an irrational fixation; it is the rational allocation of agency to whichever domain still pays for effort. The paper’s claim is not that lifting weights is bad. The claim is that an entire generation outsourcing its sense of agency to a private, paid, mirror-equipped room is symptomatic of a wider failure of the surrounding economy to provide legible effort elsewhere. The gym, in this reading, is not the default human activity it appears to be. It is a historically recent, environmentally specific coping infrastructure.

## 1. The phenomenon

A 24-year-old gets out of bed at 5:50 a.m., not because he is happy to, but because he has decided that 6 a.m. is when he lifts. He has been doing this five days a week for two years. He cannot remember the last time he won at anything else with comparable reliability. He has not been promoted. He has not been in a relationship. His best friend from college moved to a different city in 2023 and they now exchange one text per month. He does, however, deadlift 180 kg. He knows this number. He thinks about it.

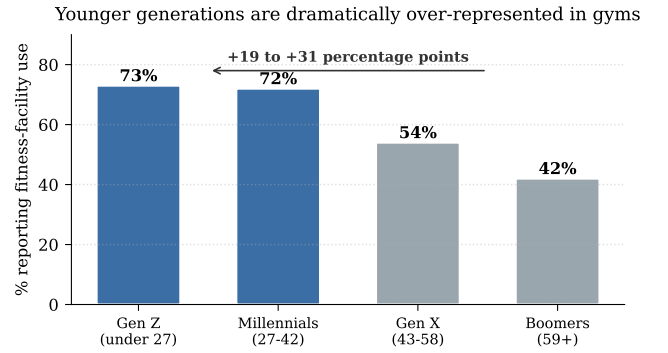
If asked why he goes, he says “I just like the routine,” which is the polite, low-information answer that protects the actual one: the gym is the only place in his week where if he shows up and tries, something measurably happens. The argument of this note is that this answer is not pathology. It is rational behavior in an environment that has stopped paying interest on effort almost everywhere else.

The puzzle is therefore not that young men go to the gym. The puzzle is that no other domain in their lives offers a comparable input-output guarantee, and that this fact deserves serious description rather than a wellness pamphlet.

## 2. The basic facts

Three sets of numbers establish the empirical landscape. First, gym participation is not evenly distributed across age. [1] report that 73% of Gen Z and 72% of Millennials use fitness facilities, against 54% of Gen X and 42% of Boomers; [2] reports that 47% of all new gym joins in 2025 came from Gen Z alone, and [3] record that the share of fitness-facility members under 25 rose from 22.9% in 2015 to 30.8% in 2024 (Figure 1).

Second, the same cohort is doing notably badly on social and emotional measures. Gallup’s 2025 analysis reports that 25% of U.S. men aged 15–34 felt lonely “a lot of the day,” against 17% of other adults; 57% report daily stress and 46% report daily worry, both markedly above older comparison groups [4]. The Survey Center



**Figure 1:** Gym participation is not flat across generations. Sources: Glofox Fall 2024 panel; ABC Fitness 2025; HFA 2025.

on American Life found that the share of U.S. men reporting *no* close friends rose from 3% in 1990 to 15% in 2021, and that 20% of single men report zero close friends [5]. These figures are summarized in Figure 2.

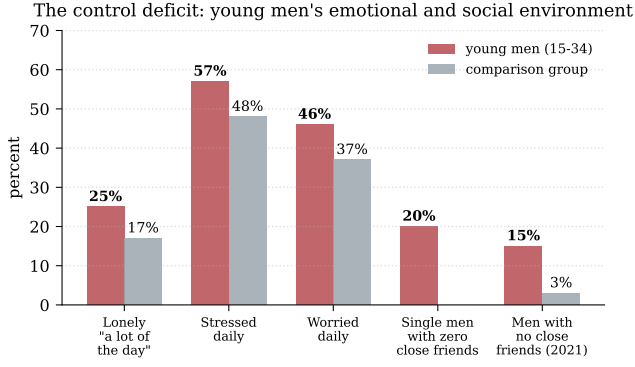
Third, the surrounding economic environment is unusually illegible for early-career workers. AI-related stress affects 65% of adults 18–34 (up from 52% one year earlier) [6]; new-graduate hiring funnels are longer and noisier; and the dating market for the same cohort is heavily mediated by platforms whose feedback is sparse. The conjunction matters: a cohort with rising gym participation is the same cohort with rising loneliness, rising AI anxiety, and a shrinking number of legible domains.

## 3. The model

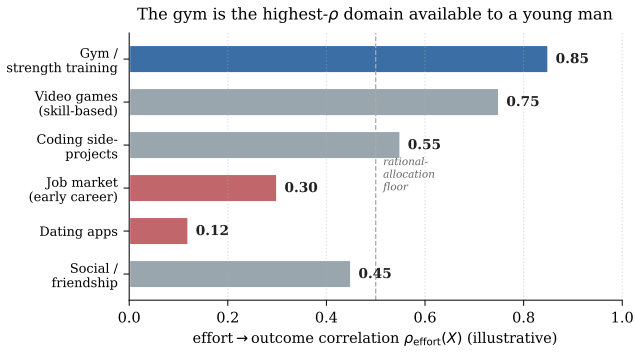
Let  $X$  index a life domain (gym, dating apps, job market, friendships, side projects). Define for each domain a behavioral coefficient

$$\rho_{\text{eff}}(X) = \text{corr}(\text{effort}, \text{outcome}),$$

the within-domain correlation between input and observable result.  $\rho_{\text{eff}}$  is bounded in  $[0, 1]$  and is the variable young men implicitly estimate when they decide where to spend the next hour.



**Figure 2:** The control deficit. Young men face a social environment in which most relational variables are moving in the wrong direction. Sources: Gallup 2025; American Perspectives Survey 2021.



**Figure 3:** Approximate effort–outcome correlations across domains. Values are illustrative, anchored to common subjective reports. The gym sits at the top because the chain effort → load → adaptation → visible result is unusually short.

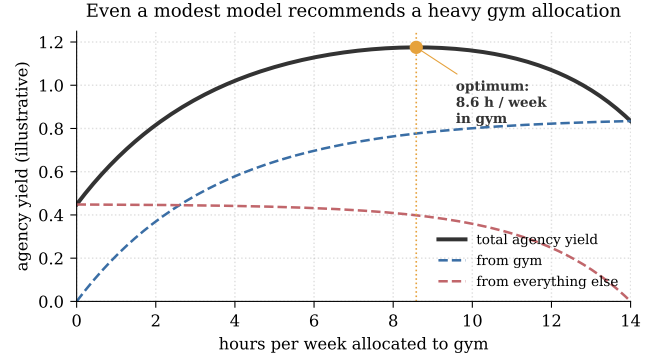
Define an actor’s *agency yield* from allocating  $h_X$  hours per week to domain  $X$  as

$$A_X(h_X) = \rho_{\text{eff}}(X) \cdot (1 - e^{-h_X/\tau_X}),$$

where  $\tau_X$  is the time scale on which the domain returns visible signal. Total agency yield is  $A = \sum_X A_X$  subject to a weekly budget  $\sum_X h_X \leq H$ . Two empirical regularities matter:

1.  $\rho_{\text{eff}}(\text{gym})$  is the highest available value in the consideration set of a typical 20–27-year-old man (Figure 3). The chain stimulus–adaptation–visible-change is short, repeatable, and almost monotonic in effort.
2.  $\tau_{\text{gym}}$  is short (weeks).  $\tau_{\text{career}}$  and  $\tau_{\text{dating}}$  are long, noisy, or undefined. The actor is rewarded for the same input on a faster cycle.

The model therefore predicts heavy gym allocation even from a fully rational agent, simply because the gym is the only term in the sum that pays interest predictably. Figure 4 solves  $\max_h A$  under a modest parameterization and recovers an interior optimum near 9 hours per week in the gym out of a 14-hour budget—roughly five sessions, which is exactly what is observed.



**Figure 4:** Even a generous parameterization of “everything else” allocates most of a young man’s discretionary budget to the gym. The optimum is not a personality flaw; it is the rational response to the  $\rho$  vector.

## 4. Worked example

**Example.** Adam, 24, holds a CS degree and lives alone. His current weekly inputs and observable outputs:

- Job applications sent: 17. Interviews received: 1. Outcome: silence.
- Hinge messages sent: 31. Matches that replied with a sentence: 4. Dates: 0.
- Hours coding a personal project: 11. External users acquired: 0.
- Gym sessions: 5. Squat 1RM at week start: 130 kg. Squat 1RM at week end: 132.5 kg.

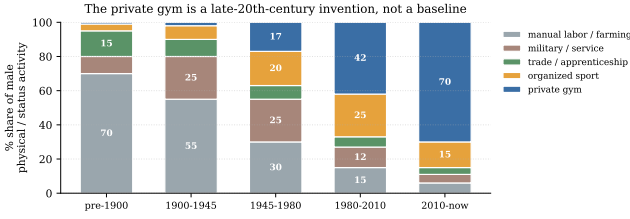
Only one of these four rows produced a positive, legible signal in the past week, and it produced it on schedule. Computing  $A_X$  under the model with  $\rho = (0.85, 0.12, 0.55, 0.30)$  for gym, dating, side-projects, and job-market respectively, more than half of Adam’s total agency yield comes from the gym—a domain into which he invested less than a third of his hours. The conclusion he draws (“the gym is the part of my life that works”) is empirically correct under the only metric he can observe in week-level resolution.

The deeper feeling reported in interviews of this cohort is not pride but *relief*. The gym is the room in which the variance of the day collapses. He puts a plate on, he picks it up. Nothing about that interaction will be reinterpreted later in light of someone else’s mood, an algorithm’s update, or a hiring freeze. The relief is what the routine is for.

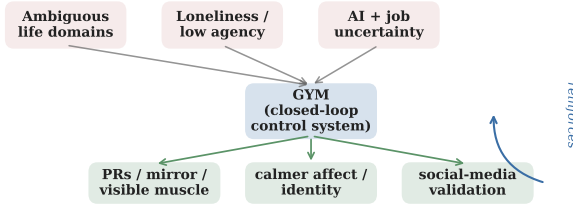
## 5. Why the gym is, historically, weird

The phrase “working out” implies a baseline against which the activity is a variation. Across most of the 20th century, that baseline did not exist. Physical exertion was bundled into work—farming, factory, military, trade—and male status was extracted from the same labor that exhausted the body. Recreational strength training in private facilities is a late-20th-century mass phenomenon, arriving with the post-industrial economy and the simultaneous decline of bodily work [8]. Figure 5 shows the rough trajectory.

This matters for the argument because it forecloses



**Figure 5:** The private gym is not a default human activity but a recent compensatory institution. Shares are schematic, drawn from the historiography of fitness culture [8], not measured.



**Figure 6:** Causes flow downward, reinforcements flow back up. The loop is not pathological; it becomes pathological only when the gym becomes the *only* domain delivering reinforcement.

the appealing reading that men have always gone to the gym and we are merely watching them “return to nature.” They have not always gone, and there is nothing especially natural about paying \$30 per month to perform purposeless work whose only legible output is a body that resembles a body that once performed purposeful work. The gym is a simulation of what previous generations were paid to do. It is also, importantly, a place that did not need to exist when the rest of the male economic environment supplied legible effort on its own. The historical claim of this paper is therefore mild: the rise of the gym is not the cause of anything. It is a fairly accurate indicator that the surrounding system has stopped supplying its previous outputs and the actor has built a private substitute.

## 6. The reinforcement loop

The mechanism by which the gym sustains itself, once entered, is a tight positive-feedback loop (Figure 6). Ambiguous life domains, social thinness, and AI-driven career uncertainty raise the marginal value of any domain with high  $\rho_{\text{eff}}$ . The gym, uniquely, returns three kinds of confirmation per session: a private one (the weight moved), an embodied one (the calmer affect for the next 6–12 hours), and a public one (the slowly visible body, amplified on social media for those who choose to share). All three feed back into the decision to return tomorrow.

## 7. Discussion

The argument is not that the gym is harmful. For most of the cohort in question, regular strength training is one of the few unambiguously protective behaviors available—against depression, anxiety, sleep dysfunction, and the metabolic consequences of sedentary

work [7]. The argument is sharper: when a behavior protects against the symptoms of a deficit elsewhere, the behavior is doing the work of a missing institution. The 20–27-year-old man who lifts five days a week is not failing at life. He is succeeding at the only legible task left on his calendar.

What he is bypassing is the messy, low- $\rho_{\text{eff}}$  work of building durable friendships, pursuing intimacy that will not respond to a protocol, and committing to a career path whose payoff he cannot see from where he stands. Those are exactly the domains in which his cohort is worst off statistically. The model does not condemn him for choosing the gym; the model predicts that he will, and asks why the alternatives have such low  $\rho$  that the choice is barely a choice.

A reasonable response is therefore not to scold gym participation, which is helpful, but to ask: what would it take to make any *other* domain in a young man’s life return effort with comparable reliability? Mentorship that responds to consistency; hiring processes that close loops; social institutions that reward showing up. The gym solved one problem because a market built the solution. The other problems remain unsolved because no market has yet been able to.

## 8. Conclusion

Men aged 20–27 train at historically unusual rates not because they have rediscovered fitness but because the gym is currently the highest- $\rho$  feedback loop available to them. The behavior is rational under the parameters; what is irrational is the environment that makes those parameters so lopsided. A society in which the most reliable place a young man can experience cause and effect is a paid room with mirrors has produced something worth describing accurately, with sympathy, and without pretending it is normal.

## References

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